Study Notes:
Roberts & Sufi (2009), “Control Rights and Capital Structure: An Empirical Investigation”

(Notes)

Contents

1 Big picture: question, core idea, and what the paper contributes

1.1 Research question

The paper asks: *Do creditor control rights materially affect corporate debt policy?* More specifically, when a firm violates a financial covenant, creditors typically obtain *acceleration* and *termination* rights. The paper studies whether this transfer of control rights causes firms to reduce net debt issuance and (eventually) leverage.

1.2 Core economic mechanism

A financial covenant violation is treated as a discrete event that shifts bargaining power and control rights toward creditors. Once creditors have these rights, they can (i) renegotiate interest rates, (ii) reduce committed credit, (iii) demand collateral, or (iv) otherwise restrict the firm's ability to borrow. Therefore, a covenant violation can operate like a *negative credit supply shock* that changes capital structure, above and beyond any smooth changes in fundamentals.

1.3 Empirical contribution in one sentence

The paper provides evidence that **net debt issuance drops sharply and persistently after covenant violations**, and this pattern is consistent with **creditor intervention** rather than merely managerial leverage rebalancing.

2 Institutional background: why covenant violations matter

2.1 Financial covenants and "control rights"

Financial covenants are contractual restrictions written on accounting ratios (e.g., leverage, interest coverage, net worth, working capital). When a covenant is violated, the borrower is in *technical default*, even if it continues making interest payments. In most credit agreements, a violation gives lenders the right to:

- accelerate repayment (declare the loan due immediately), and/or
- terminate or suspend undrawn credit commitments (reduce available liquidity).

These rights shift control to creditors *in bad states*, matching a large theoretical literature on state-contingent allocation of control rights in incomplete contracts.

2.2 Why this can affect capital structure

If creditors can credibly threaten acceleration/termination, the firm may be forced to:

- reduce borrowing (lower net debt issuance),
- alter payout/investment plans to satisfy lenders,

- substitute toward equity issuance (in principle), or
- renegotiate existing debt terms.

The paper's key empirical question is whether we observe a discrete policy shift in debt financing *at the moment control rights change*.

3 Data, sample, and variable construction (what is measured)

3.1 Sample

- Public firms from 1996–2005 (firm-quarter panel).
- Covenant violations are identified from SEC 10-K and 10-Q filings (text-based indicator).
- Financial variables are from standard accounting databases (quarterly).

3.2 Key variables and notation

Index firms by i and quarters by t.

- $D_{i,t}$: book value of total debt outstanding at quarter end.
- $A_{i,t}$: total assets at quarter end.
- **Net debt issuance (scaled):**

$$\mathrm{NDI}_{i,t} \equiv \frac{D_{i,t} - D_{i,t-1}}{A_{i,t-1}} \times 10{,}000 \quad \text{(basis points of lagged assets).}$$

- **Net equity issuance (scaled):** defined analogously using changes in book equity or equity issuance measures, scaled by $A_{i,t-1}$ (also in basis points).
- **Leverage ratio:** $\mathrm{Lev}_{i,t} \equiv \frac{D_{i,t}}{A_{i,t}} \times 10{,}000$ (reported in basis points; e.g., 2,300 bps = 0.23).
- $\mathrm{Violation}_{i,t} \in \{0, 1\}$: indicator that firm i reports a covenant violation in quarter t.

3.3 Covenant controls X

A major identification issue is that covenant violations occur when fundamentals (accounting ratios) deteriorate. To separate "fundamentals changing" from "control rights changing," the paper builds a rich set of controls based on the typical ratios used in covenant contracts. Conceptually, $X_{i,t}$ includes (at least) the following categories:

- leverage and balance-sheet strength: debt/assets, net worth/assets, net working capital/assets, cash/assets;
- profitability and coverage: EBITDA/assets, cash flow/assets, net income/assets, interest expense/assets;

- interactions motivated by covenant design (e.g., debt/assets $\times$ cash flow/assets);
- standard capital structure controls: size, tangibility, market-to-book, rating indicator.

The empirical implementation uses increasingly flexible functions of these variables (levels, interactions, higher-order polynomials, and quantile dummies).

4 Identification logic and econometric models ("all models" used in the paper)

4.1 Identification intuition: a pooled threshold discontinuity

A covenant is violated when an underlying ratio crosses an (unobserved, contract-specific) threshold. The key identifying assumption is:

> *Absent financial covenants, managers' financing policy would be continuous in the underlying accounting ratios at the covenant threshold.*

Then, any *discrete* shift in financing policy associated with $\text{Violation}_{i,t}$—after flexibly controlling for the level and changes of the underlying ratios—is attributed to creditor control rights.

4.2 Model 1: baseline firm fixed effects specification (Equation (1) in the paper)

Let the dependent variable be net debt issuance in basis points. The baseline model is:

$$\frac{D_{i,t} - D_{i,t-1}}{A_{i,t-1}} = \alpha_i + \sum_{f=1}^{4} \phi_f + \sum_{t=1996q3}^{2005q2} \delta_t + \beta_0 \,\text{Violation}_{i,t} + \beta_1 \,\text{Violation}_{i,t-1} + \Gamma^\top f(X_{i,t-1}, X_{i,t}) + \eta_{i,t}, \tag{1}$$

where:

- α_i are firm fixed effects (time-invariant firm heterogeneity);
- ϕ_f are fiscal-quarter fixed effects (seasonality);
- δ_t are calendar year-quarter fixed effects (macro shocks);
- $f(\cdot)$ is a flexible function of covenant-related ratios and standard controls;
- $\eta_{i,t}$ is the error term (standard errors clustered by firm).

Interpretation of β_1 (the key coefficient). Because $\text{Violation}_{i,t-1} = 1$ means the firm violated last quarter, β_1 captures the *post-violation* change in net debt issuance. A value of $\beta_1 = -50$ means net debt issuance falls by 50 basis points of lagged assets in the quarter after a violation.

4.3 Model 2: first-difference analog

To reduce concerns about within-firm serial correlation and slow-moving trends, the paper estimates a first-difference analog:

$$\Delta\left(\frac{D_{i,t}-D_{i,t-1}}{A_{i,t-1}}\right)=\sum_{f=1}^{4}\Delta\phi_f+\sum_t\Delta\delta_t+\beta_0\,\Delta\text{Violation}_{i,t}+\beta_1\,\Delta\text{Violation}_{i,t-1}+\Gamma^\top\Delta f(X_{i,t-1},X_{i,t})+\Delta\eta_{i,t}. \tag{2}$$

Economically, this specification asks whether *changes* in issuance jump when the violation indicator switches.

4.4 Model 3: dynamic (long-run) effects on debt issuance

To study persistence, the paper estimates versions of (1) that include multiple lags:

$$\text{NDI}_{i,t}=\alpha_i+\phi_f+\delta_t+\sum_{k=0}^{8}\beta_k\,\text{Violation}_{i,t-k}+\Gamma^\top f(X_{i,t-1},X_{i,t})+\eta_{i,t}. \tag{3}$$

4.5 Model 4: dynamic effects on leverage ratios

A parallel specification replaces the dependent variable with leverage:

$$\text{Lev}_{i,t}=\alpha_i+\phi_f+\delta_t+\sum_{k=0}^{8}\theta_k\,\text{Violation}_{i,t-k}+\Lambda^\top g(Z_{i,t-1},Z_{i,t})+u_{i,t}, \tag{4}$$

where $g(\cdot)$ contains standard leverage determinants (size, tangibility, market-to-book, profitability, etc.).

4.6 Model 5: leverage rebalancing test (interaction regressions)

To test whether the post-violation drop in debt issuance is merely mechanical "rebalancing" from high leverage, the paper estimates specifications with interactions:

$$\text{NDI}_{i,t}=\alpha_i+\phi_f+\delta_t+\rho\,\text{Lev}_{i,t-1}+\kappa\,(\text{Lev}_{i,t-1}\times\text{Violation}_{i,t-1})+\varepsilon_{i,t}, \tag{5}$$

and similarly using Debt/EBITDA in place of leverage.

4.7 Model 6: initial vs. subsequent violations

Define:

- $\text{NewViolation}_{i,t}=1$ if a violation occurs at t and there was no violation in the prior four quarters;
- $\text{SubViolation}_{i,t}=1$ if a violation occurs at t but it is not a "new" violation.

The paper estimates first-difference regressions with distributed lags:

$$\Delta\mathrm{NDI}_{i,t} = \sum_{k=0}^{8} \gamma_k \,\mathrm{NewViolation}_{i,t-k} + \sum_{k=0}^{8} \psi_k \,\mathrm{SubViolation}_{i,t-k} + \mathrm{controls} + e_{i,t}, \tag{6}$$

and also includes interaction terms to test whether subsequent violations have incremental effects immediately after an initial violation.

4.8 Model 7: cross-sectional heterogeneity in the two-quarter debt-issuance response

For each *new* violation at quarter τ, define the two-quarter change in net debt issuance:

$$\Delta^{(2)}\mathrm{NDI}_{i,\tau} \equiv \mathrm{NDI}_{i,\tau+2} - \mathrm{NDI}_{i,\tau}.$$

Then estimate:

$$\Delta^{(2)}\mathrm{NDI}_{i,\tau} = a + b^{\top} W_{i,\tau} + \epsilon_{i,\tau}, \tag{7}$$

where $W_{i,\tau}$ are firm characteristics measured at the time of the violation (cash, leverage, profitability, size, tangibility, market-to-book, rating) and an indicator for an additional violation in the interim quarter.

4.9 Model 8: probit model of creditor intervention

Using a hand-collected sample of violators, define $\mathrm{Action}_{i,\tau} \in \{0,1\}$ indicating whether creditors take some explicit action after a violation. The paper estimates probit models:

$$\Pr(\mathrm{Action}_{i,\tau} = 1 \mid Q_{i,\tau}) = \Phi(Q_{i,\tau}^{\top}\omega), \tag{8}$$

and reports marginal effects of covariates $Q_{i,\tau}$.

5 Read the evidence: figures and tables (interpretation + reproduced tables)

5.1 Table I: how common are covenant violations?

Main message. Violations are common (roughly one quarter of firms experience at least one violation over the sample). Violation probabilities are much higher than default probabilities, especially for higher-rated firms. This supports the view that covenant violations are an *early and frequent* trigger for renegotiation/control shifts, distinct from outright payment default.

Table 1: Table I: Covenant Violations

Panel A: Fraction of Firms That Violate Financial Covenant	
	Percentage of Firms Reporting Violation
Totals	
Total sample	25.6%
Firms with average book leverage ratio > 0.05	30.0%
By industry	
Agriculture, minerals, construction	28.6%
Manufacturing	25.4%
Transportation, communication, and utilities	25.2%
Trade—wholesale	34.8%
Trade—retail	23.3%
Services	24.6%
By size (book assets)	
Less than $100M	28.8%
$100M to $250M	28.8%
$250M to $500M	25.0%
$500M to $1,000M	21.7%
$1,000M to $2,500M	18.7%
$2,500M to $5,000M	17.8%
Greater than $5,000M	10.6%
Borrower does not have credit rating	26.6%
Borrower has credit rating	22.3%
Panel B: 1-Year Probabilities of Default by Credit Rating	
Credit rating	**1-year prob. of covenant violation / S&P 1-year cumulative default prob.**
A or better	1.0% / 0.0%
BBB	3.1% / 0.2%
BB	6.8% / 0.9%
B	9.4% / 7.2%
CCC or worse	18.4% / 21.9%
Unrated	10.0% / –

5.2 Table II: summary statistics (scale, dispersion, and units)

Main message. Net debt issuance is extremely dispersed (large standard deviation), with a median near zero. This motivates using within-firm variation and fixed effects. Covenant-control ratios are also dispersed, which matters for the flexible-control design.

Table 2: Table II: Summary Statistics (Firm-Quarter Panel, 1996–2005)

	Mean	**Median**	**SD**
Capital structure variables			
Net debt issuance (basis points)	50.5	0.0	400.8
Net equity issuance (basis points)	39.8	0.4	166.8
Book debt$_t$/assets$_t$	0.228	0.182	0.221
Covenant control variables			
Net worth$_t$/assets$_t$	0.495	0.518	0.287
Net working capital$_t$/assets$_t$	0.254	0.235	0.271
Cash$_t$/assets$_t$	0.199	0.092	0.231
EBITDA$_t$/assets$_{t-1}$	0.006	0.026	0.068
Cash flow$_t$/assets$_{t-1}$	−0.007	0.017	0.074
Net income$_t$/assets$_{t-1}$	−0.022	0.006	0.077
Interest expense$_t$/assets$_{t-1}$	0.005	0.003	0.006
Other control variables			
Market-to-book ratio	2.338	1.572	1.947
Tangible assets$_t$/assets$_t$	0.270	0.194	0.230
ln(assets$_t$)	4.900	4.910	2.384

5.3 Figure 1: event-time patterns (visual motivation for the regressions)

Main message. The figure shows a sharp drop in net debt issuance immediately after the violation (Panel A), no comparable increase in net equity issuance (Panel B), and a gradual decline in leverage over subsequent quarters (Panel C). This motivates the dynamic regressions and the interpretation that creditor rights affect debt capacity rather than inducing equity substitution.

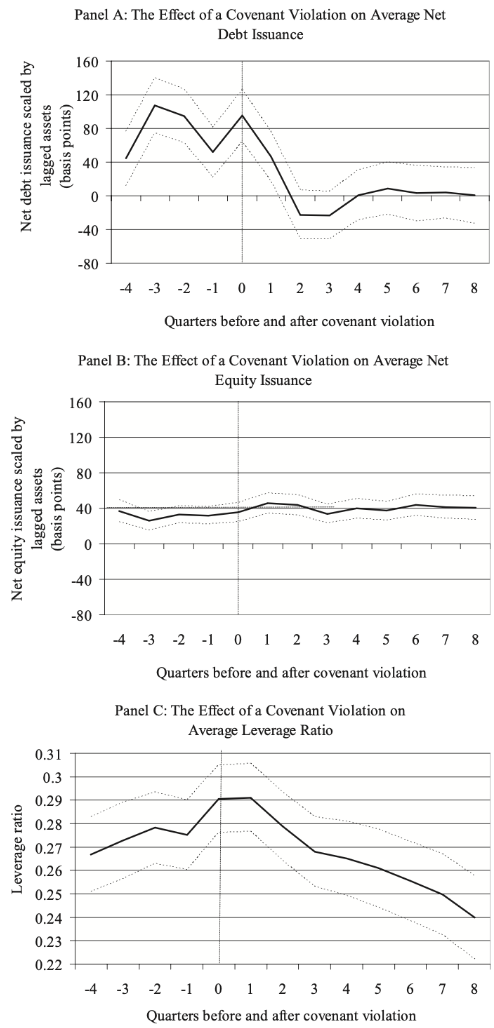

Figure 1: Figure 1: The effect of a covenant violation on capital structure (event time).

5.4 Table III: baseline causal evidence on net debt issuance

What the table estimates. Table 3 estimates (1) (Panel A) and its first-difference analog (2) (Panel B), with increasingly flexible functions of covenant controls.

How to read the coefficients. The economically relevant coefficient is on Violation_{t-1}: it measures how net debt issuance in quarter t changes *after* a violation in quarter $t-1$.

Main message. Across specifications, Violation_{t-1} is about -50 to -62 basis points and highly significant. Adding very flexible controls does not eliminate the effect, which supports the discontinuity interpretation.

Table 3: Table III: Covenant Violations and Net Debt Issuance

	Panel A: Fixed Effects			
	(1)	(2)	(3)	(4)
Dependent variable: Net debt issuance/assets$_{t-1}$ (basis points)				
Covenant violation$_t$	8.4	3.2	2.2	3.2
	(8.1)	(7.6)	(7.7)	(7.6)
Covenant violation$_{t-1}$	-62.2^{**}	-50.3^{**}	-54.3^{**}	-50.3^{**}
	(7.8)	(7.2)	(7.2)	(7.2)
Covenant controls	None	Yes	Yes + interactions	Very flexible $f(X)$
Number of firm-quarters	135,736	135,736	135,736	135,736
Number of firms	6,381	6,381	6,381	6,381
R^2	0.051	0.183	0.187	0.204
	Panel B: First Differences			
	(1)	(2)	(3)	(4)
Dependent variable: Change in net debt issuance/assets$_{t-1}$ (basis points)				
Covenant violation$_t$	9.2	-3.3	-2.3	2.2
	(11.0)	(10.1)	(10.1)	(10.0)
Covenant violation$_{t-1}$	-44.9^{**}	-60.4^{**}	-59.7^{**}	-50.3^{**}
	(11.2)	(10.3)	(10.3)	(10.3)
Covenant controls	None	Yes	Yes + interactions	Very flexible $f(X)$
Number of firm-quarters	123,557	123,557	123,557	123,557
Number of firms	6,345	6,345	6,345	6,345
R^2	0.003	0.139	0.140	0.159

** Statistically distinct from zero at the 1% level.
"Very flexible $f(X)$" corresponds to the specification with covenant controls, interactions, squares/cubes, and quantile indicators.

5.5 Table IV: long-run persistence and leverage dynamics

What the table estimates. Table 4 estimates (3) and (4) with 8 lags of the violation indicator.

Main message (debt issuance). Net debt issuance remains significantly lower for many quarters after the violation: the negative coefficients on Violation$_{t-1}$ through Violation$_{t-6}$ imply persistent reductions in borrowing.

Main message (leverage). Leverage is high around the violation (positive coefficients initially) but then gradually declines; by later horizons, the leverage response becomes statistically weak, consistent with mean reversion after a sustained reduction in borrowing.

Table 4: Table IV: Long-Run Effect of Covenant Violations

	Net debt issuance/assets$_{t-1}$		**Leverage ratio**	
	(1)	(2)	(3)	(4)
Covenant violation$_t$	19.6^{*}	11.3	290.7^{**}	214.4^{**}
	(8.7)	(8.3)	(28.8)	(28.4)
Covenant violation$_{t-1}$	-32.8^{**}	-28.8^{**}	222.9^{**}	157.7^{**}
	(8.8)	(8.3)	(24.9)	(24.6)
Covenant violation$_{t-2}$	-51.6^{**}	-43.1^{**}	128.4^{**}	106.9^{**}
	(8.5)	(8.0)	(22.8)	(22.4)
Covenant violation$_{t-3}$	-27.6^{**}	-21.8^{**}	125.3^{**}	107.0^{**}
	(8.8)	(8.3)	(22.7)	(22.4)
Covenant violation$_{t-4}$	-26.6^{**}	-22.6^{**}	56.3^{*}	43.4^{*}
	(9.0)	(8.5)	(22.3)	(21.6)
Covenant violation$_{t-5}$	-41.5^{**}	-34.0^{**}	69.3^{**}	60.2^{**}
	(8.9)	(8.4)	(22.5)	(21.9)
Covenant violation$_{t-6}$	-27.1^{**}	-25.5^{**}	40.7	27.1
	(9.1)	(8.6)	(21.7)	(21.0)
Covenant violation$_{t-7}$	-17.9^{*}	-17.2^{*}	9.4	2.1
	(8.7)	(8.2)	(23.0)	(22.4)
Covenant violation$_{t-8}$	-30.6^{**}	-33.6^{**}	-22.9	-21.8
	(9.1)	(8.6)	(27.8)	(27.0)
Controls	None	Flexible covenant controls	None	Standard leverage controls
Number of firm-quarters	92,862	92,862	92,862	92,862
Number of firms	5,654	5,654	5,654	5,654
R^2	0.110	0.215	0.790	0.798

*, ** Statistically distinct from zero at the 5% and 1% level, respectively.

5.6 Table V: economic magnitudes (how big is the effect?)

Main message (Panel A). The effect of a violation on net debt issuance is large relative to classic capital structure correlates. A violation reduces net debt issuance by about 66 basis points in the quarter after the violation.

Main message (Panel B). The long-run leverage response to a violation is economically meaningful: about -235 basis points (i.e., -2.35 percentage points of assets) in the long run.

Table 5: Table V: Economic Magnitudes

	Panel A: Net Debt Issuance		
	(1) Coefficient	(2) Within-firm SD	(3) Coef × 2 SD
Dependent variable: Net debt issuance/assets$_{t-1}$ (basis points)			
Covenant violation$_{t-1}$	-66.2^{**}		
	(7.6)		
ln(Assets$_{t-1}$)	-51.6^{**}	0.535	−55.2
	(3.6)		
(EBITDA/assets)$_{t-1}$	-455.5^{**}	0.036	−32.8
	(45.5)		
Market-to-book$_{t-1}$	9.3^{**}	1.191	22.2
	(1.1)		
(Tangible assets/assets)$_{t-1}$	154.4^{**}	0.071	21.9
	(24.5)		
Industry median leverage$_{t-1}$	-531.2^{**}	0.023	−24.4
	(59.4)		
Number of firm-quarters	135,736		
Number of firms	6,381		
R^2	0.107		

	Panel B: Leverage Ratios		
	(1) Long-run	(2) Within-firm SD	(3) LR × 2 SD
Dependent variable: Leverage ratio (basis points)			
Covenant violation	-235.2^{**}		
ln(Assets)	−36.0	0.235	−16.9
(EBITDA/assets)	$-3{,}096.9^{**}$	0.036	−223.0
Market-to-book	-62.1^{**}	0.830	−103.1
(Tangible assets/assets)	679.1^{*}	0.042	57.0
Industry median leverage	−491.6	0.010	−9.8
Number of firm-quarters	73,411		
Number of firms	5,118		
R^2	0.064		

*, ** Statistically distinct from zero at the 5% and 1% level, respectively.
In Panel B, *, ** indicate the long-run effect is statistically distinct from zero at the 5% and 1% level.

5.7 Table VI: is this just leverage rebalancing?

Main message. Even within leverage quartiles, violators issue less debt than non-violators. The interaction regressions show that the debt issuance sensitivity to leverage (or debt/EBITDA) becomes *more negative* after violations, consistent with a constraint imposed by creditor control rights rather than a purely mechanical adjustment.

Table 6: Table VI: Covenant Violations versus Leverage Rebalancing

Panel A: Leverage Ratio Quartiles		
	No violation$_{t-1}$	**Violation$_{t-1}$**
Leverage quartile 1	106	99
Leverage quartile 2	56	14^{++}
Leverage quartile 3	39	-16^{++}
Leverage quartile 4	69	-27^{++}
Panel B: Debt-to-EBITDA Quartiles		
Debt/EBITDA quartile 1	91	57
Debt/EBITDA quartile 2	40	-30^{++}
Debt/EBITDA quartile 3	25	-23^{++}
Debt/EBITDA quartile 4	32	-20^{++}

Panel C: Fixed Effects Regressions		
Dependent variable: Net debt issuance/assets$_{t-1}$ (basis points)	(1)	(2)
Leverage ratio$_{t-1}$	-500.0^{**}	
	(22.4)	
Leverage ratio$_{t-1}\times$Violation$_{t-1}$	-135.1^{**}	
	(21.6)	
Debt/EBITDA$_{t-1}$		-19.5^{**}
		(1.2)
Debt/EBITDA$_{t-1}\times$Violation$_{t-1}$		-4.0^{*}
		(1.6)
Number of firm-quarters	104,383	78,643
Number of firms	4,765	4,272
R^2	0.116	0.105

$^{++}$ Statistically distinct from "no covenant violation" at the 1% level.
*, ** Statistically distinct from zero at the 5% and 1% level, respectively.

5.8 Table VII: initial vs. subsequent violations (discrete control-rights channel)

Main message. The sharpest decline in net debt issuance occurs after *initial* violations (the first time in at least a year). Subsequent violations have small and statistically weak effects. This pattern is consistent with a discrete change in the creditor's control rights and bargaining position at the first violation.

Table 7: Table VII: Initial versus Subsequent Covenant Violations

Dependent variable: Change in [Net debt issuance/assets$_{t-1}$ (basis points)]	(1)	(2)	(3)
New violation$_t$	17.1	16.2	17.1
	(15.9)	(15.9)	(15.9)
New violation$_{t-1}$	-80.7^{**}	-81.1^{**}	-64.0^{**}
	(16.9)	(18.0)	(21.7)
New violation$_{t-1}$×Subsequent violation$_t$			−44.7
			(34.5)
New violation$_{t-2}$	-66.4^{**}	-56.4^{**}	-68.3^{**}
	(16.1)	(16.6)	(20.3)
New violation$_{t-2}$×Subsequent violation$_{t-1}$			−27.7
			(36.8)
New violation$_{t-2}$×Subsequent violation$_t$			45.7
			(40.4)
New violation$_{t-3}$	0.6	8.3	0.6
	(15.6)	(16.1)	(15.6)
New violation$_{t-4}$	−19.0	−12.2	−19.0
	(15.9)	(16.6)	(15.9)
New violation$_{t-5}$	-42.5^{**}	−30.1	-42.5^{**}
	(16.4)	(17.1)	(16.4)
New violation$_{t-6}$	−10.0	0.4	−10.1
	(16.0)	(17.1)	(16.0)
New violation$_{t-7}$	−12.4	−3.8	−12.4
	(15.8)	(17.3)	(15.8)
New violation$_{t-8}$	5.0	6.5	4.9
	(16.5)	(17.7)	(16.5)
Subsequent violation$_t$		−1.8	
		(15.0)	
Subsequent violation$_{t-1}$		−28.8	
		(17.9)	
Subsequent violation$_{t-2}$		−1.5	
		(17.9)	
Subsequent violation$_{t-3}$		−4.8	
		(17.8)	
Subsequent violation$_{t-4}$		−9.4	
		(17.5)	
Subsequent violation$_{t-5}$		−11.7	
		(17.4)	
Subsequent violation$_{t-6}$		1.2	
		(17.9)	
Subsequent violation$_{t-7}$		18.3	
		(19.4)	
Subsequent violation$_{t-8}$		−17.0	
		(15.6)	
Number of firm-quarters	88,485	88,485	88,485
Number of firms	5,609	5,609	5,609
R^2	0.131	0.131	0.131

*, ** Statistically distinct from zero at the 5% and 1% level, respectively.

5.9 Table VIII: cross-sectional heterogeneity (who is hit harder?)

Main message. The average two-quarter decline is large (negative constants). Higher leverage is associated with a larger decline. Having a credit rating is associated with a *smaller* decline (a positive coefficient), consistent with better access to alternative financing.

Table 8: Table VIII: Cross-sectional Heterogeneity of the Effect of Covenant Violation

Dependent variable: Δ[Net debt issuance/assets$_{t-1}$] over two quarters after new violation	(1)	(2)	(3)	(4)
Constant	-126.1^{**}	-106.2^{**}	-47.8	15.6
	(15.3)	(18.2)	(64.3)	(70.2)
Subsequent violation$_{t-1}$		-58.9	-52.2	-47.9
		(32.5)	(32.3)	(32.4)
Cash/assets$_{t-2}$			-49.4	-52.9
			(105.4)	(105.1)
Leverage ratio$_{t-2}$			-440.4^{**}	-483.1^{**}
			(87.9)	(89.0)
Market-to-book$_{t-2}$			29.0^{*}	28.1^{*}
			(13.2)	(13.2)
EBITDA/assets$_{t-2}$			212.8	259.0
			(316.9)	(317.8)
(Tangible assets/assets)$_{t-2}$			105.9	106.2
			(73.0)	(73.0)
ln(assets)$_{t-2}$			-0.2	-14.7
			(9.4)	(11.6)
Has S&P credit rating$_{t-2}$				114.5^{*}
				(53.1)
Number of observations	1,593	1,593	1,593	1,593
Number of firms	1,332	1,332	1,332	1,332
R^2	0.000	0.002	0.026	0.028

*, ** Statistically distinct from zero at the 5% and 1% level, respectively.
Timing: covariates with $t-2$ are measured at the quarter of the new violation; the dependent variable is the change from the violation quarter to two quarters after.

5.10 Table IX: direct evidence of creditor intervention (mechanism)

Panel A: what creditors do. In a random sample of new violators, creditors take explicit actions in about one-third of cases. When creditors take action, the subsequent change in net debt issuance is strongly negative.

Panel B: when creditors act. Creditors are more likely to act when firms have higher leverage and less financial slack (lower cash, lower EBITDA). Rated firms are less likely to face explicit creditor action, consistent with better outside options.

Table 9: Table IX: The Response of Creditors to Covenant Violations

Panel A: Response to Covenant Violations

	(1) Fraction	(2) Change in net debt issuance (conditional)
Creditor takes some action	0.322	−418
Reduction in size of credit facility	0.238	−475
Interest rate increased	0.148	−401
Additional collateral required	0.072	−436
Creditor grants waiver, no action reported	0.626	29^{++}
Borrower terminates credit agreement	0.044	−190

Panel B: Probit Estimates (Marginal Effects)

Dependent variable: Creditor takes some action following violation $\in \{0, 1\}$	(1)	(2)	(3)
Subsequent violation$_{t-1}$	0.15^{*}	0.12^{**}	0.12^{**}
	(0.04)	(0.05)	(0.05)
Cash/assets$_{t-2}$		-0.43^{**}	-0.44^{**}
		(0.16)	(0.16)
Leverage ratio$_{t-2}$		0.43^{**}	0.47^{**}
		(0.12)	(0.12)
Market-to-book$_{t-2}$		−0.03	−0.02
		(0.02)	(0.02)
EBITDA/assets$_{t-2}$		-1.09^{*}	-1.15^{*}
		(0.49)	(0.49)
(Tangible assets/assets)$_{t-2}$		−0.02	−0.02
		(0.10)	(0.10)
ln(assets)$_{t-2}$		0.00	0.03
		(0.01)	(0.02)
Has S&P credit rating$_{t-2}$			-0.16^{*}
			(0.06)
Number of observations	474	474	474
R^2	0.018	0.074	0.084

$^{++}$ Statistically distinct from "creditor takes some action" at the 1% level (Panel A comparison).

*, ** Statistically distinct from zero at the 5% and 1% level, respectively.

6 Synthesis: what we should remember for research use

6.1 Key takeaways

1. Covenant violations are frequent and distinct from outright default.
2. Debt financing falls sharply after violations; equity issuance does not offset the drop.
3. The effect is robust to very flexible controls for covenant-related fundamentals.
4. The effect is persistent and appears as a gradual decline in leverage.
5. Patterns are consistent with a **creditor control-rights / credit supply** channel.

6.2 Limitations / interpretive cautions (useful when extending)

- Covenant violations are observed from filings; some violations or creditor actions may be unreported.
- Contract-specific covenant thresholds are not directly observed; identification relies on flexible control functions.
- Results focus on public firms and may not directly generalize to private firms.